Afsaneh H.Ebrahimi

Parkville | a.hasanebrahimi@unimelb.edu.au | +61430025372 | afsaneh-hasanebrahimi

Summary

Ph.D. candidate at The University of Melbourne with a focus on trustworthy AI, including robustness, fairness, and adversarial robustness in deep learning and multimodal models. Academic background is complemented by industry experience as a data scientist and multiple teaching assistant roles.

Research Interests

- Trustworthy AI
- Robustness of multimodal models
- Quantitative Analysis

Education

The University of Melbourne — *2024 - Present*
Ph.D. in Computing and Information Systems
- **Thesis:** Towards robustness and fairness in vision-language models (VLMs)
- **Supervisors:** Prof. Christopher Leckie, Prof. Sarah M.Erfani, and Dr. Hanxun Huang

University of Tehran — *2019 - 2022*
M.Sc. in Electrical and Computer Engineering
- **GPA:** 3.88 / 4
- **Thesis:** An analysis of adversarial attacks in autonomous driving systems based on deep learning

University of Tehran — *2014 - 2019*
B.Sc. in Electrical and Computer Engineering
- **GPA:** 3.38 / 4
- **Thesis:** Design a toolbox for data fusion methods

Research Experience

PhD Graduate Researcher — Melbourne, Vic
The University of Melbourne, School of Computing and Information Systems — *Feb. 2024 - Present*
- Enhanced robustness against adversarial attacks in VLMs
- Proposed methods to mitigate bias in VLMs
- Enhanced hallucination-mitigation in VLMs

Master Student Researcher — Tehran, Iran
University of Tehran, Computational Audio-Vision Lab — *Feb. 2021 - Sep. 2022*
- Proposing a robustness method against adversarial attacks based on VAEs
- Utilizing the separation index as a metric for robustness test

Working Experience

PhD Quant Focus — Sydney, Australia
Optiver — *Jul. 2026*
- Selected as one of 16 PhD students for the Optiver PhD Quant Lab 2026, a highly selective two-day research program bringing together doctoral students to work with Optiver researchers on advanced quantitative modelling and algorithmic trading challenges.

Data Scientist (Strategy Team) — Tehran, Iran
SnappFood (Leading Online Food Ordering Company in Iran) — *Sep. 2023 - Jan. 2024*

- Designed and deployed machine learning pipelines to optimize recommender systems, leading to a 10% increase in customer engagement.
- Developed an optimized carousel that boosted user engagement, driving a 17% increase in morning orders.

Teaching Experience

Graduate Researcher Academic Associate — University of Melbourne
Course: Statistical Machine Learning — *Jul. 2025 - Present*

Academic Tutor for Elements of Data Processing — University of Melbourne
Instructors: Prof. James Bailey, Prof. Eduard Hovy — *Feb. 2025 - June 2025*

Teaching Assistant for Machine Learning — University of Tehran
Instructors: Prof. Babak N.Araabi, Dr. M.Reza Abolghasemi Dehaqani — *Sep. 2020 - June 2021*

Teaching Assistant for Deep Learning — University of Tehran
Instructors: A/Prof. Ahmad Kalhor — *Feb. 2021 - June 2021*

Teaching Assistant for Statistical Inference — University of Tehran
Instructors: Dr. Behnam Bahrak — *Feb. 2021 - Jan. 2022*

Volunteer Experience

Website Coordinator *Doctoral Colloquium 2025, The University of Melbourne* — Melbourne, VIC
Jul. 2025 – Oct. 2025
- Designed, developed, and maintained the event website.
- Coordinated registration processes and managed participant inquiries.

Check-in Assistant *AJCAI 2024 Conference* — Melbourne, VIC
Nov. 2024
- Assisted with attendee check-ins, registration verification, and provided conference information to participants.

Registration Desk *Doctoral Colloquium 2024, The University of Melbourne* — Melbourne, VIC
Oct. 2024
- Supported registration and provided logistical assistance to attendees.

Honors and Awards

Selected as one of 16 PhD students for the Optiver PhD Quant Lab 2026.	*2026*
Awarded Women in Machine Learning (WiML) travel funding.	*2026*
Awarded FEIT-Eng & IT travel scholarship.	*2026*
Awarded a travel grant to attend the Responsible AI Symposium, hosted by CSIRO and the Australian Institute for Machine Learning at the University of Adelaide.	*2025*
Selected as one of four Ph.D. mentees across Australia for the 2025 L'Oréal-UNESCO For Women in Science Mentoring Scheme.	*2025*
Awarded a top-up stipend as part of the ARC-funded Centre of Excellence in Automated Decision-Making and Society (ADM+S).	*2025*
Awarded a Melbourne Research Scholarship to pursue a Ph.D. at The University of Melbourne, Australia.	*2024*
Ranked 3^{rd} among MSc Students Entered in 2019 at the University of Tehran, School of Electrical and Computer Engineering, Electrical Engineering.	*2022*
Ranked 12^{th} in the MSc Entrance Exam among more than 14000 Participants.	*2019*
Top 0.1%, Nationwide Entrance Exam of Iranian Universities, a very competitive exam with more than 220,000 participants.	*2014*

Talks

Bias and Fairness in Vision-Language Models ADM+S 2025 Symposium
The University of Queensland, Brisbane, Australia *Jul. 2025*

The Importance of Adversarial Robustness in Vision-Language Models ADM+S 2024 Symposium
University of New South Wales, Sydney, Australia *Oct. 2024*

Publications

- **A. Hasanebrahimi**, H. Huang, C. Leckie, and S. Erfani, "Density-Aware Translation of Spurious Correlations in Zero-Shot VLMs," International Conference on Machine Learning (ICML), 2026.
- **A. Hasanebrahimi**, H. Huang, C. Leckie, J. Bailey, and S. Erfani, "GeoDetect: Geometric Adversarial Detection for VLPs," European Conference on Computer Vision (ECCV), 2026.
- **A. Hasanebrahimi**, H. Huang, C. Leckie, and S. Erfani, "VisER: Visual Evidence and Reliance for Object Hallucination Detection in LVLMs," under review at EMNLP 2026.
- **A. Hasanebrahimi**, B. K. Baghbaderani, R. Hosseini, and A. Kalhor, "Density Estimation Helps Adversarial Robustness," International Conference on Computer and Knowledge Engineering (ICCKE), 2023.
- B. K. Baghbaderani, **A. Hasanebrahimi**, A. Kalhor and R. Hosseini, "Adversarial Robustness Evaluation with Separation Index," International Conference on Computer and Knowledge Engineering (ICCKE) 2023.

Technical Skills

Programming: Python, R, C, MATLAB, Bash Scripting, CUDA

Machine Learning Frameworks: Pytorch, Tensorflow, Keras, Scikit-learn, SciPy, Pandas, Numpy

Microcontrollers: AVR, ARM, Arduino

Others: Git, MySQL
Hardware Description Language (HDL): Verilog, Modelsim, Quartus

Soft Skills

- Adaptability
- Teamwork
- Time Management

Important Courses

- Neural network and Deep learning (Grade: 4/4)
- Pattern Recognition and Machine Learning (Grade: 4/4)
- Data Analytics (Grade: 4/4)
- Statistical Inference (Grade: 4/4)
- Linear Algebra (Grade: 4/4)

References

Available Upon Request